SALTDRAG · DESIGN DECISION RECORD

SaltDrag TRL 2 review decisions

Design Molecule Lab · designmolecule.com

DOCUMENT CONTROL

Document ID	SDG-DDR-001	Version	0.1
Status	Draft	Date	2026-10-03
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/saltdrag		
Source	docs/decisions/0001-trl2-review-decisions.md @ c7f0938		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-03	Amish Chadha	TRL 2 review decisions, made under Amish's 2026-10-03 pre-approval of the batch

0001: TRL 2 review decisions

- **Date:** 2026-10-03
- **Status:** accepted. Decided by Amish under his pre-approval of 2026-10-03: "start with the first 14 repos from the list of 29 projects. I pre-approve the batch runs along with any recommendations you come up with. I also accept any cost overruns or variations from the assumed scope cost."

Context

The scaffold (TRL 1, 2026-09-30) left the concept open on how the capstans stand and are anchored, the drive, the rope, the rake build and the people who would host a trial. Taking the repo to TRL 2 and on to TRL 3 in one run needed each of these settled. Amish pre-approved every recommendation for this batch, so each choice below is recorded as decided; safety choices take the conservative option and say what evidence would relax them.

Options considered and decisions

Table 1. Decisions at the TRL 2 review.

#	Question	Options	Decision	Reason
D1	Kind of capstan	Warping capstan with a person tailing the rope; geared drum winch that stores the rope	Geared drum winch, still called a hand capstan	Nobody holds a loaded rope by hand; simpler and safer for a small crew (conservative)
D2	Where the capstan stands	On the bund crest; on dry ground behind the bund with a roller on the crest	Behind the bund, drum 1.5 m behind the crest centre, with a bund roller saddle on the crest	A 0.4 m crest cannot carry a capstan and two people cranking; the roller keeps the rope off the bund
D3	Drive	Worm gear; two-stage spur; single-stage spur	Single spur stage, module 4, 84 and 14 teeth (6 to 1), two 330 mm cranks set 180 degrees apart	Efficient, repairable, gears can be bought cut in India; meets R2 with margin
D4	Rope	Nylon; polypropylene; polyester; high-modulus polyethylene	12 mm polyester double braid, 45 m per side	Low stretch limits snap-back energy; UV and brine tolerant; affordable (nylon stores too much energy; polypropylene degrades in sun)
D5	Rake	Bolted sections; one piece; timber	One-piece 3.0 m FRP box beam with a 300 mm UHMW-PE blade	Light (12.8 kg beam), does not rust, carried by two people; width set by the 3 kN design drag
D6	Blade height	Fixed; adjustable skids	Three UHMW-PE skids on slotted stainless hangers; blade 10 mm above the floor, adjustable 5 to 25 mm	Protects the hand-hardened floor (R4)
D7	Anchoring	Stakes; deadweight; screw-in ground anchors	Two screw-in helical anchors per capstan, chained to ears on the side frames at rope height	Holds 3 kN without ballast; quick to move; chain at rope height stops tipping
D8	Materials	Painted steel; galvanised steel; stainless throughout	Hot-dip galvanised steel on dry ground; FRP, UHMW-PE and 316 stainless in the pan; stainless-insert thermoplastic bearings; daily freshwater rinse	Nothing that rusts sits in brine (R7) at a moderate cost
D9	Holding and paying out (safety)	Pawl only; brake only; pawl and brake	Pawl always engaged while winding; paying out only on the band brake with the crank handles removed	Conservative: a released crank cannot spin back. Relaxing it (cranks left on while paying out) would need a self-sustaining brake or a no-back crank tested with CalRig
D10	Snap-back protection (safety)	Rope choice only; dampers and exclusion zone	Low-stretch rope plus a weighted damper bag on each loaded rope, and a 2 m exclusion zone along the rope line	Conservative; a test of rope behaviour at break with CalRig would show whether the zone can be narrowed
D11	Anchor proof load (safety)	Trust the estimate; proof-load on site	Every anchor proof-loaded on site to 1.5 times its share of the rated pull before first use	Soil behind the bunds is unknown; a site pull test is the only evidence
D12	Design pan	20 m; 30 m; 45 m	30 m between bund crests, rope for 45 m	Covers the pans described as tens of metres wide; survey to confirm

#	Question	Options	Decision	Reason
D13	Crew	Three; four	Four: two cranking, one at the far capstan, one moving rollers and ropes	Keeps cranking shifts short in the heat
D14	Co-design partner	Producer organisations and NGOs working with Agariyas	First candidate to approach: SEWA (Self Employed Women's Association), which works with salt-farming families in the Little Rann	Women-led and already working on livelihoods in the pans. Not agreed
D15	First trial region	Little Rann of Kutch; coastal Kutch creeks	Little Rann of Kutch, first candidate	Where the problem was found and the users are
D16	Machine-use condition	Assume hand tools are allowed; ask first	Ask the forest department in writing, through the partner, before any trial	The registration conditions forbid machine use; a written answer removes the doubt
D17	Shared hand capstan block	Separate designs; share	SaltDrag's capstan offered as the first candidate for the common block with SiltHaul	One proven capstan for both; SiltHaul is not changed here

Consequences

- The concept gains a bund roller saddle (BOM 12) and loses the windrow board; ballast bags follow from the lift-off calculation (SDG-DDR-002).
- R10 to R12 are added to SDG-REQ-001 for rope safety, holding and rope capacity.
- The site facts the decisions rest on (crest width, ground behind the bunds, soil strength, drag per metre) are listed in SDG-DEC-001 to be confirmed.
- `budget_usd` in `project.yaml` is unchanged; the estimated cost is reported against it as a value-engineering target.